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# Thermodynamic assessment of the Ytterbium–Germanium binary system

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**Abstract** The Ytterbium–Germanium binary system was evaluated using the CALPHAD method through Thermo-Calc software package; the thermodynamic parameters involved in the Gibbs free energy models of the phases are optimized based on all available literature experimental data including the phase equilibria information and few thermodynamic measurements. The Ytterbium–Germanium system contain six intermetallic compounds $Yb_2Ge$, $Yb_5Ge_3$, $Yb_5Ge_4$, $Yb_{11}Ge_{10}$, $Yb_3Ge_5$, and $Yb_3Ge_8$, which are all treated as stoichiometric phases. The excess term of the Gibbs free energy of the liquid phase was assessed with the recent exponential temperature dependence of the interaction energies by Kaptay and compared with the linear temperature dependence of Redlich–Kister polynomial equation results. The Germanium (Diamond_A4) and the two allotropic transformations of Ytterbium: βYb (fcc_A1) and γYb (bcc_A2), were kept as pure element phases since no solubility of Germanium in Ytterbium and vice versa. A reasonable agreement between the calculated results (the phase diagram and model parameters) and literature data was obtained for this system. The optimization process allows us to accomplish the results which are reported in a series of figures and tables.

**Electronic supplementary material** The online version of this article (doi:10.1007/s10973-017-6203-x) contains supplementary material, which is available to authorized users.

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**Keywords** Thermodynamic assessment · Phase diagram · CALPHAD method · Yb–Ge binary system · Redlich–Kister and Kaptay equations

## Introduction

The Germanium element has been the focus of considerable applications; it is used in telecommunication sector as a component of glass in the fiber optics (the addition of Germanium to glass increases its index of refraction and confines the light in the fiber). The interesting physical property as an intrinsic semiconductor makes it an important candidate in electronic applications as a transistor element. Germanium is also employed in infrared optical devices, such as infrared spectroscopes and infrared detectors, because of being transparent to infrared radiation.

In the field of alloying materials, the addition of Germanium to some other elements such as tin and Aluminum-Magnesium alloys, improves their physical strength and hardness, (the addition of a meager 0.35% Germanium to tin results in a material with double the hardness) [1]. In fluorescent lamps, Germanium is combined as a phosphor and as a polymerization catalyst for polyethylene terephthalate (PET). Additionally, the photovoltaic industry is concerned as a high-demand sector for Germanium. On the other side, Rare Earth Elements (REEs) have been increasingly used in high-performance functional materials because of their magnetic, optical, thermal, and electrical properties. These properties make the REEs indispensable in a wide range of advanced materials, such as permanent magnets, magnetic refrigerants, hydrogen storage materials [2–4], new energy technologies, and national security applications [5]. It is known that the valence electron

configuration for the REEs and specially lanthanides is trivalent $4f^n(5d6s)^3$, except Eu and Yb elements, which has a divalent $4f^{n-1}(5d6s)^2$ configuration [6], and this configuration gives them considerable chemical and metallurgical properties. Consequently, the compounds originating from the Rare Earth Elements (REEs) tetrelides (Si, Ge, … group 14 elements), for example, Ytterbium–Germanium, have unique structures and unusual interesting physical properties such as giant magnetocaloric effect, magnetostriction, and magnetoresistance [7]. These characteristics make the REEs-Ge alloys useful in many fields of technology, such as electric contact materials in Ge-waver-based electronic devices [8], as well in telecommunication because of sharply structured photoluminescence bands from rare earth intra-4f-shell transitions [9].

Moreover, many ternary systems RE–TM–X (RE = Rare Earth, TM = Transition Metal, X = Si or Ge) lead to the formation of intermetallic compounds which are known by a number of outstanding properties such as superconductivity, heavy-fermion behaviors, Kondo behavior, intermediate valency, and magnetic properties which has been widely described and discussed [10–13].

In order to give a better understanding of the constitutional properties and potential technological applications of the Ge–RE alloys materials, it is important to investigate the phase diagrams and develop a thermodynamic database of these systems based on the CALPHAD method. The present study is a part of the thermodynamic investigation of the Ge–RE (RE = Yttrium) systems already calculated in our laboratory by Idbenali and Servant [14].

For the Yb–Ge system, the experimental information was obtained from the literature concerning phase equilibria and thermodynamic properties. Up to now, no thermodynamic assessment of this system has been reported. The scope of the present work is to compile the experimental information and to study its consistency in comparison with the calculated results obtained by using CALPHAD approach.

## Review of literature data

### Equilibrium diagram

The Ytterbium–Germanium binary system has been investigated by different scientists. It was firstly studied by Eremenko et al. [15]. These authors reported the existence of three intermediate phases: $Ge_{10}Yb_{11}$ ($Ho_{11}Ge_{10}$-type), $Ge_{5-x}Yb_3$ with two structural modifications ($Th_3Pd_5$-type and $Mn_5Si_3$-type), they also mentioned the existence of a large liquid miscibility gap extending from 45 to 90 at.% Yb. Those results were compiled by Massalski et al. [16] in the book Binary Alloy Phase Diagrams. In 2003, Pani and Palenzona [17] reinvestigated the Ytterbium–Germanium binary system under ambient pressure and over the whole compositional range of Germanium based on the results from differential thermal analysis (DTA), X-ray diffraction (XRD), optical microscopy (LOM), and Electron Probe MicroAnalysis (EMPA); the result is shown in Fig. 1. These authors brought many changes to complete the Ytterbium–Germanium phase diagram: First they refuted the existence of the earlier reported miscibility gap [15]; second they confirmed the existence and crystal structure of four phases, namely $Yb_2Ge$ ($Cl_2Pb$ type), $Yb_5Ge_3$ ($Mn_5Si_3$ type), $Yb_{11}Ge_{10}$ ($Ho_{11}Ge_{10}$), and $Yb_3Ge_5$ ($Th_3Pd_5$); third they obtained two new intermediate phases, namely the $Yb_5Ge_4$ characterized with the orthorhombic $Sm_5Ge_4$-type, and $Yb_3Ge_8$ with an unusual novel triclinic type; and finally they annulled the existence of $Yb_2Ge_3$ (AlB type) which was the first compound discovered in the Ytterbium–Germanium system reported in 1964 by Gladyshevskii et al. [18].

The Ytterbium–Germanium binary system includes the following phases: liquid; pure elements γ-Yb (bcc_A2), β-Yb (fcc_A1), and Ge (diamond_A4); and six stoichiometric compounds [16]: $Yb_5Ge_4$ melt congruently at 1640 °C. $Yb_2Ge$, $Yb_5Ge_3$, $Yb_{11}Ge_{10}$, $Yb_3Ge_5$, and $Yb_3Ge_8$ formed by peritectic reactions at 1205, 1585, 1500, 1075, and 865 °C, respectively. The addition of Germanium to

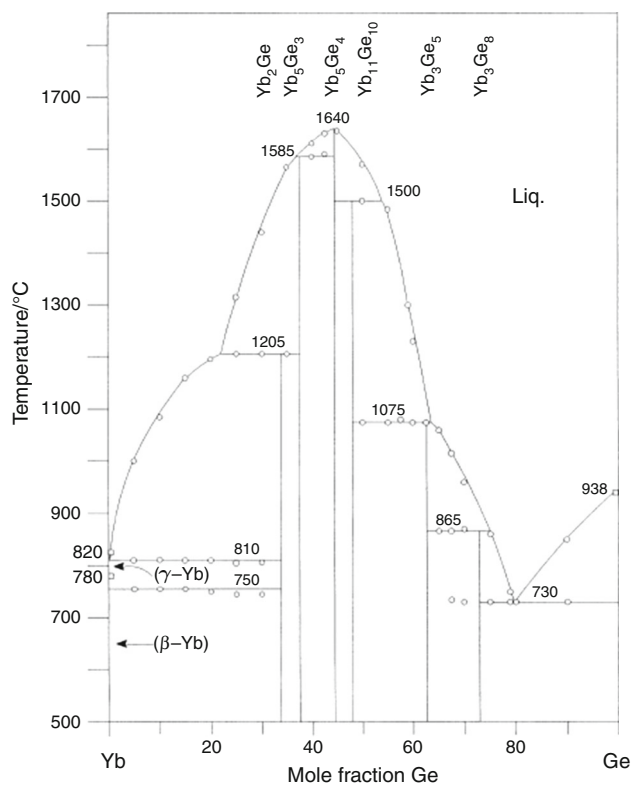


**Fig. 1** Experimental phase diagram of the Ytterbium–Germanium binary system according to Pani and Palenzona [17]

**Table 1** Crystal data of the intermetallic phases in the Yb–Ge system [17]

| Compound | at.% Ge | Crystal system | Structure type | Space group | Pearson code |
|---|---|---|---|---|---|
| $Yb_2Ge$ | 33.3 | Orthorhombic | $PbCl_2$ | *Pnma* | oP12 |
| $Yb_5Ge_3$ | 37.5 | Hexagonal | $Mn_5Si_3$ | *P6₃/mcm* | hP16 |
| $Yb_5Ge_4$ | 44.4 | Orthorhombic | $Sm_5Ge_4$ | *Pnma* | oP36 |
| $Yb_{11}Ge_{10}$ | 47.6 | Tetragonal | $Ho_{11}Ge_{10}$ | *I4/mmm* | tI84 |
| $Yb_3Ge_5$ | 62.5 | Hexagonal | $Th_3Pd_5$ | *P62m* | hP8 |
| $Yb_3Ge_8$ | 72.7 | Triclinic | $Yb_3Ge_8$ | *P1* | aP22 |

Ytterbium results in eutectic reaction among liquid, Yb (bcc_A2), and $Yb_2Ge$ at 810 °C and <1 at.% Ge. In Ge-rich part, the addition of Ytterbium to Germanium results in a eutectic reaction among liquid, $Yb_3Ge_8$, and Ge (diamond_A4) at 730 °C and about 80 at.% Ge, in accordance with Eremenko [15].

Several groups of authors [19, 20] contributed to the identification of a new compound $YbGe_3$ in Ytterbium–Germanium binary system. The reaction of Yb and Ge in liquid indium was studied by Peter Sebastian et al. [19]. Using the energy-dispersive X-ray spectroscopy (EDS) measurements, they observed a few small triangular-shaped single crystals of a new binary $YbGe_3$ compound, which was subsequently determined to have the composition of $YbGe_{2.83}$.

Using high-pressure and high-temperature reactions, Harada et al. [20] obtained two new Ytterbium Germanides, $Yb_3Ge_5$, and $YbGe_3$. $Yb_3Ge_5$ new orthorhombic phase (crystallizes in the $Pu_3Pd_5$ structure) was obtained by annealing hexagonal $Yb_3Ge_5$ (the ambient pressure phase) at 13 GPa and 1000 °C for 1 h; $YbGe_3$ (crystallizes in the $Cu_3Au$ structure) was synthesized under pressures higher than 2 GPa in temperatures between 800 and 1200 °C after mixing Ge with $Yb_3Ge_8$. $YbGe_3$ is a new $Cu_3Au$-type compound with metallic properties [20].

### Thermodynamic data

The only investigation of the heats of formation of five out of six intermediate phases in the Ytterbium–Germanium binary system was reported for the first time by Balducci et al. [7]. The following values were found for the various compounds: $Yb_2Ge$, $-66.2 \pm 8.3$ kJ mol$^{-1}$ at.$^{-1}$; $Yb_5Ge_3$, $-69.7 \pm 8.2$ kJ mol$^{-1}$ at.$^{-1}$; $Yb_5Ge_4$, $-73.8 \pm 8.2$ kJ mol$^{-1}$ at.$^{-1}$; $Yb_{11}Ge_{10}$, $-69.9 \pm 8.2$ kJ mol$^{-1}$ at.$^{-1}$; and $Yb_3Ge_5$, $-49.8 \pm 13.2$ kJ mol$^{-1}$ at.$^{-1}$. These results were calculated by Balducci et al. [7] from the high-temperature decomposition reactions of Ytterbium–Germanium intermediate phases using Knudsen effusion mass spectrometry (KEMS) and Knudsen effusion mass loss (KEWL) techniques after calculating the reaction enthalpies derived from measuring the Yb(g) decomposition pressures as a function of temperature. They also investigated the stability of $Yb_3Ge_5$(s) compound by DFT-LCAO-B3LYP (density functional theory linear combination of atomic orbitals-hybrid b3lyp exchange–correlation functional), and they used first principles calculations for deriving its enthalpy of formation at 0 K. The value of the enthalpy of formation for this compound is $-55.9 \pm 6.2$ kJ mol$^{-1}$ at.$^{-1}$.

The other available experimental data are the heat capacity of $Yb_5Ge_4$ measured by Ahn et al. [21] at low temperature using an adiabatic heat pulse calorimeter in d.c magnetic fields ranging below 7 T. This result cannot be used in the optimization as SGTE data [22] that are referred to 298 K and valid only above this temperature. The experimental data measured by Ahn et al. [21] concerning the heat capacity of $Yb_5Ge_4$ compound were used by Balducci et al. [7] to calculate the entropy of formation of $Yb_5Ge_4$ at 298 K, $\Delta_f S_{298}(Yb_5Ge_4(s)) = -3.26 \pm 0.20$ J K$^{-1}$ mol$^{-1}$ at.$^{-1}$; then, the entropy of formation of $Yb_{11}Ge_{10}$ was derived: $\Delta_f S_{298}(Yb_{11}Ge_{10}(s)) = -1.0 \pm 0.8$ J K$^{-1}$ mol$^{-1}$ at.$^{-1}$. Nikolaenko et al. [23] have measured the integral enthalpies of mixing of liquid alloy for $x(Ge) = 0.16$ at 1223 K. All these experimental data were used to evaluate the thermodynamic parameters in the present modeling.

### Crystallographic data

The crystallographic data for the intermediate phases formed in the Ytterbium–Germanium binary system are reported in Table 1.

## Procedure of thermodynamic modeling

The first thing to do before starting the optimization process is consulting the literature search. It is very important to collect and categorize all the data which are able to contribute in the optimization of the studied system. The second step involves the application of CALPHAD method, using the PARROT module in the Thermo-Calc software developed by Jansson [24] and Sundman et al. [25], to optimize the model parameters of the Gibbs energy

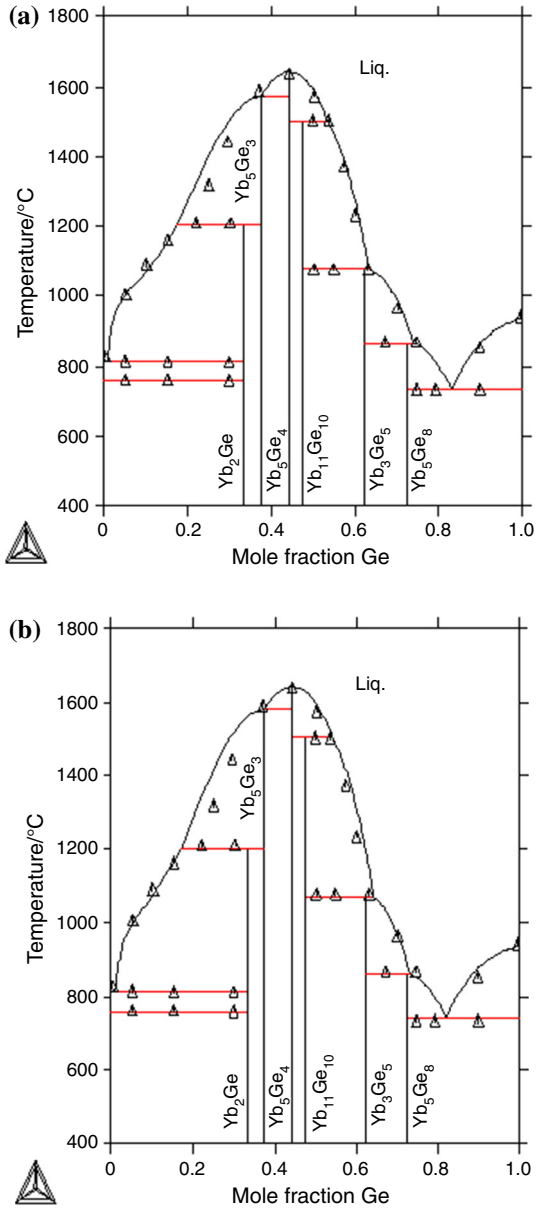


**Fig. 2** **a** Calculated Ytterbium–Germanium phase diagram using the thermodynamic parameters optimized with the Redlich–Kister model [26] compared with the experimental data of Pani and Palenzona [17]. **b** Calculated Ytterbium–Germanium phase diagram using the thermodynamic parameters optimized with the Kaptay model compared with the experimental data of Pani and Palenzona [17]

expressions of each phase in the system. This step can be divided to two sub-steps.

The first is the choice of thermodynamic models for the Gibbs free energy of the individual phases. The second is adjusting the parameters in order to minimize the square sum of the errors between the experimental data and the computed values.

The modeling procedure was carried out as follows; the first stage of the optimization began with introducing a file of the stability of the liquid phase. The liquid was treated first as a substitutional solution using Redlich–Kister polynomial [26], and then, Kaptay model [27–29] was used in order to avoid the occurrence of an unwanted inverted miscibility gap in the solution phases during the calculation of Ytterbium–Germanium phase diagram. In the case of the compounds, the congruent intermetallic compound $Yb_5Ge_4$ was taken into account first. Then, the other intermetallic phases were introduced in the modeling one by one. Next, experimental data related to invariant equilibria were added to the calculation. Finally, all parameters are adjusted again and assessed simultaneously in the final calculation steps until most of experimental data were accounted for within the expected uncertainty limits. The optimization of the parameters is completed as soon as the global error is reduced to the minimum.

## Thermodynamic models

Thermo-Calc uses different thermodynamic formalisms following the structure of the phase to be modeled.

### Pure elements

The Gibbs free energy function for the pure element $i(i = \mathrm{Yb},\ \mathrm{Ge})$ in its stable state of the phase $\Phi(\Phi = \mathrm{bcc_A2},\ \mathrm{fcc_A1},\ \text{and}\ \mathrm{Diamond_A4})$ is described by an equation of the following form:

$$G^{\Phi} - H^{\Phi}_{\mathrm{SER}} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{\Phi} \quad (1)$$

where $H_{\mathrm{i}}^{\mathrm{SER}}(298.15\ \mathrm{K})$ is the molar enthalpy of the element $i$ at 298.15 K and 1 bar in its standard element

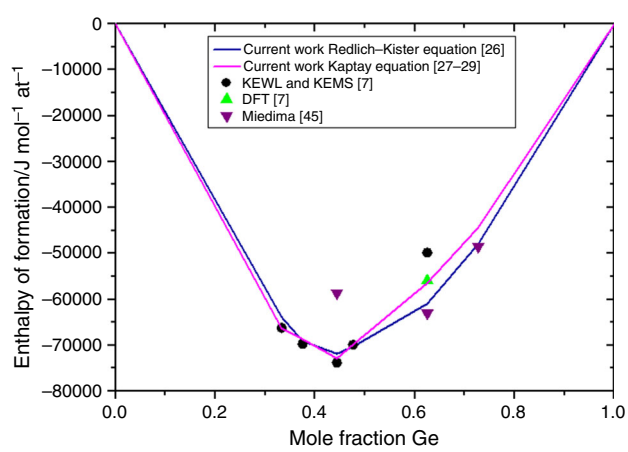


**Fig. 3** Comparison between the calculated enthalpies of the six intermetallic compounds using Redlich–Kister equation [26] (*blue line*) and Kaptay equation [27–29] (*pink line*), with the experimental data from equilibrium vapor pressure techniques (●) [7], predicted with the Miedema model (▼) [45] and obtained by DFT calculation (▲) [7]. (Color figure online)

**Table 2** Comparison between the enthalpies and entropies of formation of the intermetallic compounds of the Ytterbium–Germanium system derived from the present optimization and taken from literature

| Intermetallic compounds | Composition/ at.% Ge | Enthalpy of formation/ kJ mol$^{-1}$ at.$^{-1}$ | Technique used | Entropy of formation/ J mol$^{-1}$at.$^{-1}$ K$^{-1}$ | References |
|---|---|---|---|---|---|
| $Yb_2Ge$ | 33.3 | −66.2 ± 8.3 | KEWL and KEMS | – | [7] |
| | | −63.856 | Optimization[R–K] | −5.4 | Current work |
| | | −66.303 | Optimization[K] | −8 | Current work |
| $Yb_5Ge_3$ | 37.5 | −69.7 ± 8.2 | KEWL and KEMS | – | [7] |
| | | −69.100 | Optimization[R–K] | −4.8 | Current work |
| | | −68.649 | Optimization[K] | −5.8 | Current work |
| $Yb_5Ge_4$ | 44.4 | −73.8 ± 8.2 | KEWL and KEMS | −3.26 ± 0.2 | [7] |
| | | −58.6 | Prediction | – | [45] |
| | | −71.836 | Optimization[R–K] | −3 | Current work |
| | | −72.846 | Optimization[K] | −4.7 | Current work |
| $Yb_{11}Ge_{10}$ | 47.6 | −69.9 ± 8.2 | KEWL and KEMS | −1 ± 0.8 | [7] |
| | | −70.319 | Optimization[R–K] | −1.99 | Current work |
| | | −69.8555 | Optimization[K] | −3 | Current work |
| $Yb_3Ge_5$ | 62.5 | −49.8 ± 13.2 | KEWL and KEMS | – | [7] |
| | | −63.0 | Prediction | – | [45] |
| | | −55.9 ± 6.2 | DFT | – | [7] |
| | | −60.952 | Optimization[R–K] | −1.3 | Current work |
| | | −56.426 | Optimization[K] | −0.13 | Current work |
| $Yb_3Ge_8$ | 72.7 | −48.5 | Prediction | – | [45] |
| | | −47.941 | Optimization[R–K] | −0.84 | Current work |
| | | −44.343 | Optimization[K] | −0.005 | Current work |

Equilibrium vapor pressure techniques: *KEWL* Knudsen effusion mass loss and *KEMS* Knudsen effusion mass spectrometry

*DFT* density functional theory

[R–K] Redlich–Kister equation [26]

[K] Kaptay equation [27–29]

reference (SER) state fcc_A1, bcc_A2 for Ytterbium and diamond_A4 for Germanium, and $T$ is the absolute temperature; $a$, $b$, $c$, …, $h$ are the coefficients. The Gibbs free energy of the element i, in its SER state, is denoted by *GHSERi* as follows:

$$\begin{aligned}\mathrm{GHSER_{Ge}} &= {}^{0}G_{\mathrm{Ge}}^{\mathrm{Diamond_A4}} \\ &= G_{\mathrm{Ge}}^{\mathrm{Diamond_A4}} - H_{\mathrm{Ge}}^{\mathrm{Diamond_A4}}(298.15\ \mathrm{K})\end{aligned} \quad (2)$$

$$\mathrm{GHSER_{Yb}} = {}^{0}G_{\mathrm{Yb}}^{\mathrm{bcc_A2}} = G_{\mathrm{Yb}}^{\mathrm{bcc_A2}} - H_{\mathrm{Yb}}^{\mathrm{bcc_A2}}(298.15\ \mathrm{K}) \quad (3)$$

In the present modeling, the Gibbs energies functions are taken from the SGTE (Scientific Group Thermo data Europe) database compiled by Dinsdale [30].

## Solution phases

The molar Gibbs energy of the liquid phase can be expressed as the sum of the reference part, the ideal part, and the excess part:

$$G_{\mathrm{m}}^{\mathrm{liquid}} = {}^{\mathrm{ref}}G^{\mathrm{liquid}} + {}^{\mathrm{id}}G^{\mathrm{liquid}} + {}^{\mathrm{ex}}G^{\mathrm{liquid}} \quad (4)$$

$$\begin{aligned}G_{\mathrm{m}}^{\mathrm{liquid}} = {} & x_{\mathrm{Yb}}\,{}^{0}G_{\mathrm{Yb}}^{\mathrm{liquid}} + x_{\mathrm{Ge}}\,{}^{0}G_{\mathrm{Ge}}^{\mathrm{liquid}} + RT(x_{\mathrm{Yb}}\ln x_{\mathrm{Yb}} \\ & + x_{\mathrm{Ge}}\ln x_{\mathrm{Ge}}) + {}^{\mathrm{ex}}G_{\mathrm{m}}^{\mathrm{liquid}}\end{aligned} \quad (5)$$

where $R$ is the ideal gas constant, $T$ is the temperature in Kelvin, $x_{\mathrm{Yb}}$ and $x_{\mathrm{Ge}}$ are the mole fraction of the elements Yb and Ge, respectively.

The excess Gibbs energy ${}^{\mathrm{ex}}G_{\mathrm{m}}^{\mathrm{liquid}}$ is described by the Redlich–Kister polynomial [26] as follows:

$${}^{\mathrm{ex}}G_{\mathrm{m}}^{\mathrm{liquid}} = x_{\mathrm{Yb}}x_{\mathrm{Ge}}\sum_{i=0}^{n}{}^{\mathrm{i}}L_{\mathrm{Yb,Ge}}^{\mathrm{liquid}}(x_{\mathrm{Yb}} - x_{\mathrm{Ge}})^{\mathrm{i}} \quad (6)$$

The $i$th interaction parameters between components Yb and Ge ${}^{\mathrm{i}}L_{\mathrm{Ge,Yb}}^{\mathrm{liquid}}$ were temperature-dependent and are taken as a linear function of temperature: Redlich–Kister equation [26] as follows:

$${}^{\mathrm{i}}L_{\mathrm{Ge,Yb}}^{\mathrm{liquid}} = a_{\mathrm{i}} + b_{\mathrm{i}}T \quad (i = 0, 1, 2, \ldots) \quad (7)$$

With the excess entropy $s = -b_{\mathrm{i}}$.

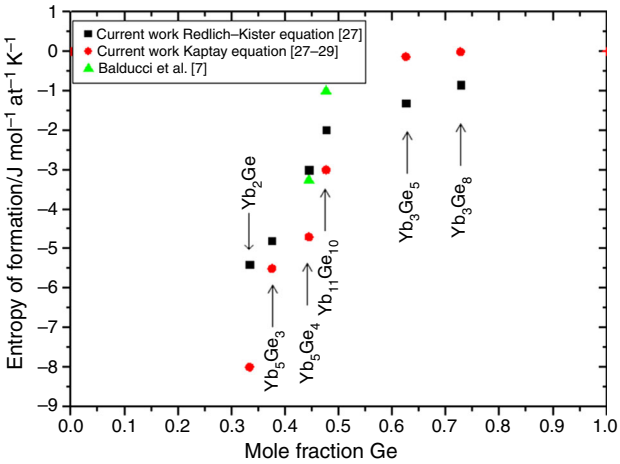


**Fig. 4** Calculated entropy of formation of the intermetallic compounds calculated with the Redlich–Kister (■) and Kaptay (●) models compared with the experimental data (▲) [7]. (The reference states of pure elements of Yb and Ge are bcc_A2 and diamond_A4 phases, respectively.)

In fact, during the CALPHAD procedure, it was found by Chen et al. [46] that the often used linear dependence of temperature of the excess Gibbs energy of the solution phases may lead to an unwanted inverted miscibility gap at high temperature in many calculated phase diagrams. In order to deal with this problem, Kaptay [27–29] suggested a new function to describe the excess Gibbs energy of the solution phases by assuming that the excess Gibbs energy should tend toward 0 at very high temperatures. This form is taken as an exponential function of temperature:

$$ {}^{i}L^{\text{liquid}}_{\text{Ge,Yb}} = h^{i}_{0} \exp\left(-\frac{T}{\tau^{i}_{0}}\right) \quad (i = 0, 1, 2, \ldots) \tag{8} $$

with $h^{i}_{0}$ the enthalpy part (J mol$^{-1}$) of ${}^{i}L^{\text{liquid}}_{\text{Ge,Yb}}$ at $T = 0$ K; the $\tau^{i}_{0} = \frac{h^{i}_{0}}{s^{i}_{0}}$ parameter must be always positive. By identifying $h^{i}_{0}$ with $a_i$ and $\tau^{i}_{0}$ with $a_i/b_i$, Eq. (8) becomes Eq. (9) with the excess entropy $s = -b_i$.

$$ {}^{i}L^{\text{liquid}}_{\text{Ge,Yb}} = a_i \exp\left(\left(-\frac{b_i}{a_i}\right) * T\right) \tag{9} $$

where the constants $a_i$ and $b_i$ are to be optimized in the present work.

This equation is an answer to the previous problem fact that we always verified during the optimization and calculation of different systems. This function has been applied successfully to the thermodynamic optimization of the Ni–Sc [31], Fe–Y [32], Si–Ta [33], Fe–Hf [34], Zn–Zr [35], Ru–Zr [36], Ru–Ti [37], Cr–Ge [38], Mg–Si [39], Al–Er [40], Gd–Tl [41], Mg–Zn, Mg–Nd, Mg–Zn–Nd [42], Co–Gd [43], and Zr–Si–N [44] systems. Neither high-temperature nor low-temperature artifacts appeared in these 15 fully optimized systems.

No excess term of the Gibbs energy was optimized for terminal solid solutions, because no solubility of Ge in Yb and Yb in Ge was reported.

### Stoichiometric compounds

In the Ytterbium–Germanium binary system, all intermediate compounds $Yb_2Ge$, $Yb_5Ge_3$, $Yb_5Ge_4$, $Yb_{11}Ge_{10}$, $Yb_3Ge_5$, and $Yb_3Ge_8$ are treated as stoichiometric phases and described as $A_pB_q$.

The Gibbs energy of each phase is denoted as ${}^{0}G_{\text{Yb}_p\text{Ge}_q}$ and described by the following formula:

$$ {}^{0}G_{\text{Yb}_p\text{Ge}_q} = \frac{p}{p+q}\,{}^{0}G_{\text{Yb}} + \frac{q}{p+q}\,{}^{0}G_{\text{Ge}} + a + bT \tag{10} $$

where ${}^{0}G_{\text{Yb}}$ and ${}^{0}G_{\text{Ge}}$ are the Gibbs energy of the pure elements ytterbium and germanium, respectively; $a$ and $b$ are optimized for each of the compounds in the present work.

The coefficients $a$ can be estimated according to the corresponding experimental enthalpies of formation and then evaluated by taking into account both phase equilibria and thermodynamic data.

## Results and discussion

The computed phase diagram of the Ytterbium–Germanium system using the present set of thermodynamic parameters is illustrated in Fig. 2a, b. In Fig. 2a, we compare our calculated phase diagram, using the Redlich–Kister model (linear temperature dependence of the excess term of the liquid phase [26]), with the literature data [17]. In Fig. 2b, our calculated phase diagram with the Kaptay model (exponential temperature dependence of the excess term of the Gibbs energy of the liquid phases [27–29]) is compared with the same literature data. We noted that our calculated values concerning invariant reactions of four intermetallic compounds, $Yb_5Ge_3$, $Yb_{11}Ge_{10}$, $Yb_3Ge_5$, and $Yb_3Ge_8$, which are formed by peritectic reactions in this system, are in good agreement with the temperatures and compositions determined experimentally by Pani and Palenzona [17].

However, one exception that is noted for the $Yb_2Ge$ phase which is formed through the following peritectic reaction $Yb_2Ge \leftrightarrow Liq + Yb_5Ge_3$ in (17.0 at.% Ge) shows a small disagreement with experimental results (22.0 at.% Ge). We tried to minimize this difference and to obtain a better description of the liquid in the Yb-rich part by using the exponential temperature dependence given by

**Table 3** Optimized thermodynamic parameters of the Ytterbium–Germanium system using Redlich–Kister equation [26]

| Phases | Thermodynamic model | Redlich–Kister equation [26] |
|---|---|---|
| Liquid | (Yb, Ge) | ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Yb,Ge}} = -270,129.67 + 51.83 * T$ |
| | | ${}^{1}L^{\mathrm{Liq}}_{\mathrm{Yb,Ge}} = -32,723.43$ |
| | | ${}^{2}L^{\mathrm{Liq}}_{\mathrm{Yb,Ge}} = 48,767.87$ |
| FCC_A1 | $(\mathrm{Yb, Ge})_1(\mathrm{Va})^*{}_1$ | No excess term optimized |
| BCC_A2 | $(\mathrm{Yb, Ge})_1(\mathrm{Va})^*{}_3$ | No excess term optimized |
| Diamond_A4 | (Ge) | No excess term optimized |
| $Yb_2Ge$ | $Yb_{0.667}$:$Ge_{0.333}$ | $G^{\mathrm{Yb_2Ge}}_{\mathrm{Yb,Ge}} - 0.667^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.333^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.667^{*}\mathrm{GHSERYb} + 0.333^{*}\mathrm{GHSERGe} - 63,856.95 + 5.49^{*}T$ |
| $Yb_5Ge_3$ | $Yb_{0.625}$:$Ge_{0.375}$ | $G^{\mathrm{Yb_5Ge_3}}_{\mathrm{Yb,Ge}} - 0.625^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.375^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.625^{*}\mathrm{GHSERYb} + 0.375^{*}\mathrm{GHSERGe} - 69,100 + 4.8^{*}T$ |
| $Yb_5Ge_4$ | $Yb_{0.556}$:$Ge_{0.444}$ | $G^{\mathrm{Yb_5Ge_4}}_{\mathrm{Yb,Ge}} - 0.556^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.444^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.556^{*}\mathrm{GHSERYb} + 0.444^{*}\mathrm{GHSERGe} - 71,836.47 + 3.01^{*}T$ |
| $Yb_{11}Ge_{10}$ | $Yb_{0.524}$:$Ge_{0.476}$ | $G^{\mathrm{Yb_{11}Ge_{10}}}_{\mathrm{Yb,Ge}} - 0.524^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.476^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.524^{*}\mathrm{GHSERYb} + 0.476^{*}\mathrm{GHSERGe} - 70,319.79 + 1.99^{*}T$ |
| $Yb_3Ge_5$ | $Yb_{0.375}$:$Ge_{0.625}$ | $G^{\mathrm{Yb_3Ge_5}}_{\mathrm{Yb,Ge}} - 0.375^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.625^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.375^{*}\mathrm{GHSERYb} + 0.625^{*}\mathrm{GHSERGe} - 60,952.85 + 1.33^{*}T$ |
| $Yb_3Ge_8$ | $Yb_{0.273}$:$Ge_{0.727}$ | $G^{\mathrm{Yb_3Ge_8}}_{\mathrm{Yb,Ge}} - 0.273^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.727^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.273^{*}\mathrm{GHSERYb} + 0.727^{*}\mathrm{GHSERGe} - 47,941.79 + 0.84^{*}T$ |

* ($V_a$) for vacancy

**Table 4** Optimized thermodynamic parameters of the Ytterbium–Germanium system using Kaptay equation [27–29]

| Phases | Thermodynamic model | Kaptay equation [27–29] |
|---|---|---|
| Liquid | (Yb, Ge) | ${}^{0}L^{\mathrm{Liq}}_{\mathrm{Yb,Ge}} = -289,027.55\exp(-3.07\mathrm{E}-04^{*}T)$ |
| | | ${}^{1}L^{\mathrm{Liq}}_{\mathrm{Yb,Ge}} = -24,408.82\exp(-5.04\mathrm{E}-05^{*}T)$ |
| | | ${}^{2}L^{\mathrm{Liq}}_{\mathrm{Yb,Ge}} = 51,161.12$ |
| FCC_A1 | $(\mathrm{Yb, Ge})_1(\mathrm{Va})^*{}_1$ | No excess term optimized |
| BCC_A2 | $(\mathrm{Yb, Ge})_1(\mathrm{Va})^*{}_3$ | No excess term optimized |
| Diamond_A4 | (Ge) | No excess term optimized |
| $Yb_2Ge$ | $Yb_{0.667}$:$Ge_{0.333}$ | $G^{\mathrm{Yb_2Ge}}_{\mathrm{Yb,Ge}} - 0.667^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.333^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.667^{*}\mathrm{GHSERYb} + 0.333^{*}\mathrm{GHSERGe} - 66,303.61 + 8.07^{*}T$ |
| $Yb_5Ge_3$ | $Yb_{0.625}$:$Ge_{0.375}$ | $G^{\mathrm{Yb_5Ge_3}}_{\mathrm{Yb,Ge}} - 0.625^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.375^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.625^{*}\mathrm{GHSERYb} + 0.375^{*}\mathrm{GHSERGe} - 68,649.37 + 5.51^{*}T$ |
| $Yb_5Ge_4$ | $Yb_{0.556}$:$Ge_{0.444}$ | $G^{\mathrm{Yb_5Ge_4}}_{\mathrm{Yb,Ge}} - 0.556^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.444^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.556^{*}\mathrm{GHSERYb} + 0.444^{*}\mathrm{GHSERGe} - 72,846.56 + 4.76^{*}T$ |
| $Yb_{11}Ge_{10}$ | $Yb_{0.524}$:$Ge_{0.476}$ | $G^{\mathrm{Yb_{11}Ge_{10}}}_{\mathrm{Yb,Ge}} - 0.524^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.476^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.524^{*}\mathrm{GHSERYb} + 0.476^{*}\mathrm{GHSERGe} - 69,855.6 + 3.06^{*}T$ |
| $Yb_3Ge_5$ | $Yb_{0.375}$:$Ge_{0.625}$ | $G^{\mathrm{Yb_3Ge_5}}_{\mathrm{Yb,Ge}} - 0.375^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.625^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.375^{*}\mathrm{GHSERYb} + 0.625^{*}\mathrm{GHSERGe} - 56,426.71 + 0.13^{*}T$ |
| $Yb_3Ge_8$ | $Yb_{0.273}$:$Ge_{0.727}$ | $G^{\mathrm{Yb_3Ge_8}}_{\mathrm{Yb,Ge}} - 0.273^{*298.15}H^{\mathrm{BCC_A2}}_{\mathrm{Yb}} - 0.727^{*298.15}H^{\mathrm{DIAMOND_A4}}_{\mathrm{Ge}} = 0.273^{*}\mathrm{GHSERYb} + 0.727^{*}\mathrm{GHSERGe} - 44,343.64 + 0.005^{*}T$ |

* ($V_a$) for vacancy

**Table 5** Comparison between experimental and calculated data (using Kaptay and Redlich–Kister equations) relative to the Ytterbium–Germanium invariant reactions

| Reactions | Reaction type | Literature [17] | | This work Redlich–Kister [26] | | This work Kaptay [27–29] | |
|---|---|---|---|---|---|---|---|
| | | *T*/°C | *x* liq/at.%Ge | *T*/°C | ' liq/at.%Ge | *T*/°C | *x* liq/at.%Ge |
| L ↔ γYb | Melting | 820 | 0.00 | 820 | 0.00 | 820 | 0.00 |
| γYb ↔ βYb | Allotropic | 780 | 0.00 | 780 | 0.00 | 780 | 0.00 |
| L ↔ γLa + $Yb_2Ge$ | Eutectic | 810 | 0.0007 | 814 | 0.008 | 813 | 0.009 |
| $Yb_2Ge$ ↔ L + $Yb_5Ge_3$ | Peritectic | 1205 | 0.221 | 1207 | 0.176 | 1203 | 0.171 |
| $Yb_5Ge_3$ ↔ L + $Yb_5Ge_4$ | Peritectic | 1585 | 0.371 | 1572 | 0.374 | 1581 | 0.374 |
| L ↔ $Yb_5Ge_4$ | Congruent | 1640 | 0.444 | 1643 | 0.444 | 1642 | 0.444 |
| $Yb_{11}Ge_{10}$ ↔ L + $Yb_5Ge_4$ | Peritectic | 1500 | 0.536 | 1502 | 0.534 | 1505 | 0.541 |
| $Yb_3Ge_5$ ↔ L + $Yb_{11}Ge_{10}$ | Peritectic | 1075 | 0.633 | 1078 | 0.634 | 1069 | 0.643 |
| $Yb_3Ge_8$ ↔ L + $Yb_3Ge_5$ | Peritectic | 865 | 0.746 | 866 | 0.741 | 858 | 0.733 |
| L ↔ $Yb_3Ge_8$ + Ge | Eutectic | 730 | 0.794 | 734 | 0.831 | 741 | 0.820 |
| L ↔ Ge | Melting | 938 | 1.00 | 938 | 1.00 | 938 | 1.00 |

the Kaptay formalism. However, the two phase diagrams obtained in this work using the two formalisms are similar.

As mentioned previously, the enthalpies of formation of intermetallic compounds are calculated by measuring the composition of the vapor phases during the high-temperature processes by Balducci et al. [7]. A summary of calculated enthalpies and entropies of formation of Ytterbium–Germanium intermediate phases that derived from the present optimization is represented together with the experimental results from [7], predicted values by the Miedema model [45] and the data obtained by DFT [7] for $Yb_3Ge_5$, in Fig. 3 and Table 2. As can be seen, the enthalpy of formation of $Yb_5Ge_4$ is the most negative, which is in accordance with the results obtained by Balducci et al. [7]. A satisfactory agreement is noted between the calculated and the experimental values determined for the five studied compounds [7]. The predicted value relating to $Yb_5Ge_4$ is less negative than the calculated result and the experimental one, whereas the predicted values for $Yb_3Ge_5$ and $Yb_3Ge_8$ are in agreement with our calculated results. We note that the calculated enthalpy of formation concerning $Yb_3Ge_5$ phase is in accordance with DFT value. The calculated entropies of formation at 298.15 K based on the present assessed thermodynamic parameters are plotted in Fig. 4; they are in good agreement with the data from [7]. The enthalpies part of formation is identical in the two formalisms, but the entropies part is slightly higher in the Kaptay model. For the two optimizations, the agreement is satisfactory for both entropies and enthalpies of formation.

The optimized thermodynamic parameters for all the phases in the Ytterbium–Germanium binary system obtained using the two formalisms of temperature dependence of the excess terms in the Gibbs free energy are given in Tables 3 and 4. Table 5 lists the calculated compositions and temperatures of the invariant equilibria for this binary system using two models: Redlich–Kister and Kaptay, and the results obtained by the two formalisms are nearly similar. A satisfactory agreement is achieved between experimental and computed invariant temperatures and compositions for eutectic reactions, congruent, and incongruent melting points of $Yb_2Ge$, $Yb_5Ge_3$, $Yb_{11}Ge_{10}$, $Yb_3Ge_5$, and $Yb_3Ge_8$ except the peritectic reaction mentioned before (which present a discrepancy of about 5.0 at.% Ge).

In order to check the reliability of our optimized parameters, we calculated the phase diagram when the liquid phase is suspended as indicated by different authors [46, 47]. Figure 5a (Redlich–Kister formalism) and Fig. 5b (Kaptay formalism) show that when the liquid phase is not considered during the calculation of the phase diagram, the stoichiometric phases disappear at high temperatures and only the bcc_A2 solid solution is calculated on the whole germanium composition range. Indeed, the primary phase bcc_A2 is the most stable. The difference between the two graphs of phase diagram calculated for both optimizations was due to the slightly higher values of the entropy part in the Kaptay procedure. The calculated integral enthalpy of mixing of the liquid Ytterbium–Germanium alloys referring to liquid components at 1223 K is plotted in Fig. 6 compared with a few experimental data [23]. The assessed integral enthalpies of liquid agree with the experimental data.

In Fig. 7a, b, the Gibbs free energy of mixing of the liquid phase was calculated at different temperatures: 2500–4500 K with Redlich–Kister and Kaptay equations, respectively. These two figures show that the Gibbs energies of mixing for the liquid phase calculated with linear and exponential models have remarkably similar behaviors: They both become less negative when the temperature

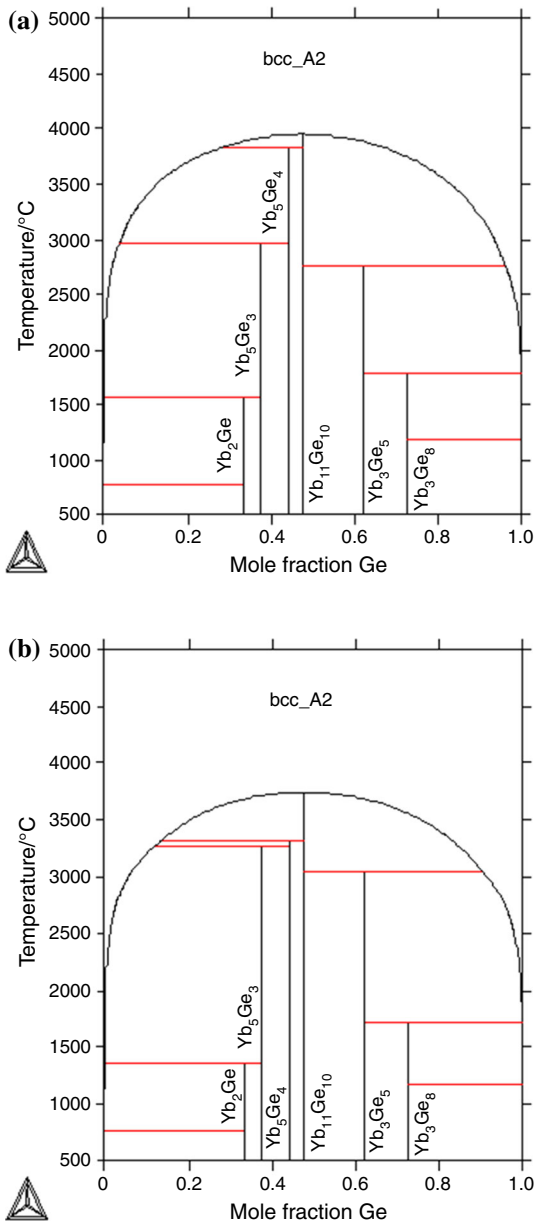


**Fig. 5** **a** Calculated Ytterbium–Germanium phase diagram with the Redlich–Kister equation [26] when the liquid phase is suspended. **b** Calculated Ytterbium–Germanium phase diagram with the Kaptay equation [27–29] when the liquid phase is suspended

increases. The difference between the two models is that in the Kaptay model the liquid is stable at very high temperatures, no spinodal points were detected until a temperature of 10,000 K, whereas in the case of the Redlich–Kister model, at 8000 K the liquid become instable. Although this behavior, we mention that no formation of an inverted miscibility gap was detected at very high temperatures, in the region of validity of parameters, during the optimization by using the two models.

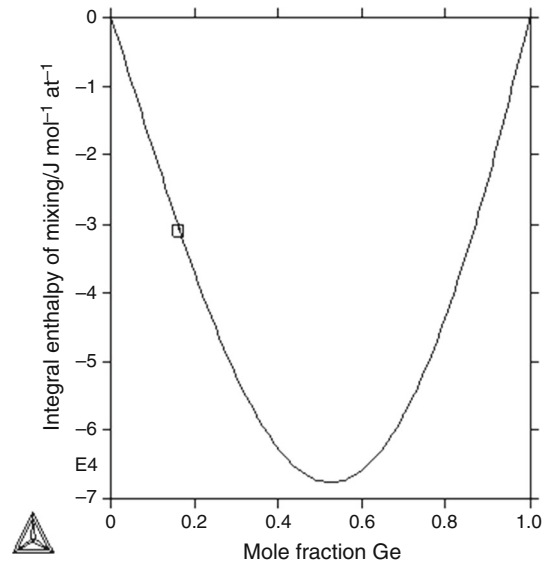


**Fig. 6** Calculated integral enthalpies of formation of the liquid phase (*continuous line*) with the experimental data measured by Nikolaenko and Turchanin [23] (*open square*) at $T = 1223$ K

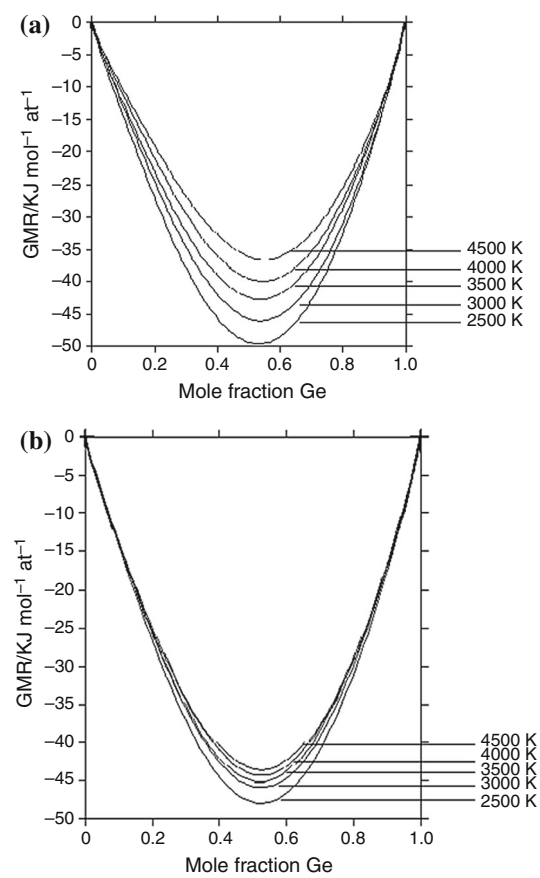


**Fig. 7** **a** Calculated Gibbs free energy of mixing of the liquid phase at different temperatures (2500–4500 K) using the Redlich–Kister model [26]. **b** Calculated Gibbs free energy of mixing of the liquid phase at different temperatures (2500–4500 K) using the Kaptay model [27–29]

## Conclusions

The experimental phase equilibria and the thermodynamic data relative to the Ytterbium–Germanium system available in the literature have been critically evaluated by means of the CALPHAD approach, in which the Gibbs free energy functions relative to the liquid phase were described by the Redlich–Kister polynomial and Kaptay formalism. In both cases, a set of self-consistent thermodynamic parameters was obtained; the calculated phase diagram and computed values are similar and in reasonable agreement with literature data. It is also worth noting that no unwanted inverted miscibility gap was formed in the liquid phase in the region of validity of parameters. Because of the lack of the information about the enthalpies of mixing of the liquid phase and the less determination of the enthalpies of formation of the compounds, the thermodynamic modeling still waiting for more experimental determination to improve and to be more exact.

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